- Operates continuously
- Production output is higher

3.5.5 Transfer Printing

Transfer printing means shifting of a design from one surface to another. In this method, designs made of pigments in paraffin or thermoplastic base can be transferred by heat and pressure to the fabric surface. The fabrics printed by this method become stiff and they are not fast to washing and light. A more effective and easier method of transferring the design intact from paper to fabric is by vaporizing the pigments in the design. Vaporizing can be done by dry heat transfer and wet heat transfer.

Disperse dyes are the only dyes which can be sublimated and used for heat transfer printing. Hence transfer printing is suitable for fabrics which have affinity to disperse dyes. Polyester, nylon and acrylics can be printed by this method. The fabric to be printed is passed through a heat transfer printing machine which brings paper and fabric together and passes them through the machine at about 204°C. Under high temperature, the dye on the printed paper sublimates and is transferred onto the fabric.



Figure 3.16 Transfer Printing

Advantages

- Production cost is low because there is no requirement of post printing treatments
- There is no wastage of pigment as the design made on paper is transferred to fabric
- Prints with excellent line details, intricacy and shading can be done
- Design can be made with many colours
- Rich and brilliant shades can be generated on woven and knitted fabrics
- Process is simple and does not require skilled labour
- High quality prints
- Less time
- Environment friendly process
- Economical for short runs

Disadvantages

- Selection of dyes is limited
- Poor colourfastness
- Not suitable for all types of fabrics

3.5.6 Batik Printing

Batik printing is a hand printing method. This is a resist style of printing. In this method, designs are made using wax and the fabric is immersed in the dye bath to colour the unwaxed portions. The wax is applied using various tools such as brushes, tjap and tjanting.

Tjanting is a spouted tool used to draw designs on the fabric with melted wax. Tjap is a pattern made of fine copper strips. The tjap is pressed on liquid wax and applied to the fabric. Wax is applied on both sides of the fabrics. After application of wax, the fabric is dyed to obtain desired colour. Only the portions not covered by the wax will absorb the dye. After the dye has been fixed, the fabric is dried, then boiled and rinsed to remove the wax. It is a slow process.

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